

Stated differently, a \$200,000 mortgage, amortized at a rate of 5%, will be paid off in exactly 20 years, when the annual payments add up to \$16,000.

Okay, you say to yourself, if you can earn a consistent 5% every year in real terms, your money will last for exactly 20 years. That's plenty of time, right?

Well, maybe, or maybe not. Remember, statistics tell us that, using moderate estimates, a 65-year-old man has a 46% chance of living for 20 more years; a 65-year-old woman has a 56% chance of living for 20 more years. So let's put two and two together and see what happens. If they earn 5%, they will run out of money in 20 years. That much is clear. But there's a 46% chance (56% for women) of living for 20 more years. In other words, there is a 46% (56%) chance of outliving your money, if you earn 5% each and every year. Why? Well, the odds of outliving your money are the odds of being alive when the money runs out. If you know exactly when the money will run out, and you know the odds of living to that point, put them together and you have the odds of outliving your money.

Similarly, if your \$200,000 nest egg earns a fixed 4% in real terms, you will run out of money even sooner—in about 18 years—because you are earning less. And the odds of living for 18 more years are, not surprisingly, higher than the odds of living for 20 more years. The chances are 54% for men and 63% for women. So if you earn 4% every year, the odds of running out of money are 54% and 63%, respectively.

Another way of saying this is that slightly less than one of every two men (and six of every ten women) will outlive \$200,000 invested at 4%, if your annual pretax needs are \$16,000.

Again, here's the procedure: Simply compute when the money will run out and then look at the probability of being alive at that time. The higher the number, the more likely it is that your standard of living is simply not sustainable.

The same formula applies in the other direction. If your capital base can earn 6% in real terms, you won't run out of money for 24 years. For a 65-year-old, 24 years might seem far out in the future. Indeed, men at that age have only a 28% chance of living that much longer (and running out of money); women have a 39% chance. These